

Class-wide second-cumulant off-diagonal stability: the Bogoliubov band

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

T-019 reframed the off-diagonal residual onto two already-tracked objects: the A-dependent condensate bubble (R_{lead} -controlled) and the beyond-mean-field two-loop = SC-SCOPE. This note closes the first one CLASS-WIDE at second cumulant. Math428 verified the off-diagonal Bogoliubov band positive for five ENUMERATED readings; here the verdict is extended to every admissible crystallographic-shell competitor through the exact band criterion and the T-018 class bound on R_{lead} . Scope: second-cumulant off-diagonal stability at the operating endpoint $I = 2 \times 10^{-3}$, anchor $\mu^2 = 5 \times 10^{-3}$. The third cumulant (SC-SCOPE) is separate; the competitor-class definition is an operator item.

2. The band-positivity criterion

From Math428 the condensate off-diagonal free-energy share is

$$\Delta F_{\text{est}}(A) = \Delta F_{\text{diag}}(A) + \rho \left[-\frac{1}{4} \sum_{Q \neq 0} W(Q)^2 B(|Q|) \right], \quad W(Q) \propto A^2,$$

with ρ the exact/second-order resummation ratio. The Bogoliubov band is positive iff $\Delta F_{\text{est}} > 0$, i.e. $\Delta F_{\text{diag}} > \rho \cdot \frac{1}{4} \sum W^2 B$:

$$\Delta F_{\text{est}} > 0 \iff \rho R_{\text{lead}} < 1, \quad R_{\text{lead}} = \frac{\frac{1}{4} \sum_{Q \neq 0} W^2 B}{\Delta F_{\text{diag}}}$$

the band sign is governed by the single product ρR_{lead} (verified: the band proxy $1 - \rho R_{\text{lead}}$ is > 0 at product 0.5, < 0 at 1.5, $= 0$ at 1).

3. At $\rho = 1$: class-wide reduction to $R_{\text{lead}} < 1$

The conservative no-resummation-credit value $\rho = 1$ (explicitly reported by Math428) gives the criterion $R_{\text{lead}} < 1$. T-018 pins R_{lead} over the admissible class via the floor identity

$$R_{\text{lead}}(Q) = \text{const} (1 + K_{\text{floor}}(Q)) I, \quad \text{const} = \frac{9}{4} \frac{u_{\text{eff}}^2 B_{\text{max}} N}{c_{\text{diag}}} = 23.2, \quad K_{\text{floor}}(Q) \leq T'(Q) \leq 13,$$

so for EVERY admissible competitor

$$R_{\text{lead}}(Q) \leq 23.2 (1 + 13) (2 \times 10^{-3}) = 0.650 < 1 \quad (\text{margin} \times 1.54).$$

Therefore $\rho R_{\text{lead}} \leq 1 \cdot 0.650 < 1$ at $\rho = 1$: the second-cumulant off-diagonal bands are POSITIVE class-wide. This extends Math428's five enumerated readings (band min $+6.7 \times 10^{-4} > 0$ at $\rho = 1$) to all admissible patterns by the same criterion.

4. The screening resummation is a credit ($\rho \leq 1$)

The resummation ratio is $\rho = 1/(1 + g)$ with $g \geq 0$ the screened-response coupling of the stable disordered parent (screening, not anti-screening). Hence $\rho \leq 1$ and the $\rho = 1$ bound is conservative. With the Math428-consistent $g = 1.03$, $\rho = 0.493$ and $\rho R_{\text{lead}} \leq 0.32$ – a further $\times 3$ margin. The class-wide stability does not rely on the resummation credit; $\rho = 1$ already closes it.

5. Status and honest residual

T4 STRONG EVIDENCE. The second-cumulant off-diagonal Bogoliubov stability is established CLASS-WIDE: $\rho R_{\text{lead}} \leq 0.650 < 1$ for every admissible crystallographic competitor (at $\rho = 1$; the screening $\rho \leq 1$ improves it). **OPEN (separate residuals):** (i) the beyond-second-cumulant third cumulant = SC-SCOPE (sunset cap $\times 1.13$, lifted@thin-certified) – thin but distinct; (ii) the competitor-class definition (crystallographic-shell subsets up to n_{pack}) – an operator/modeling item. Anchored: $u_{\text{eff}} > 0$, $g \geq 0$ (ROBUSTNESS-MU2, $[\times 0.5, \times 2]$ closed). No tier flip: B2 T6, B1 T6 on {H-LAYER}.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “ $\rho \leq 1$ is assumed, not proved class-wide.” MITIGATED: the load-bearing statement uses $\rho = 1$ (Math428's explicit conservative value), where the criterion is just $R_{\text{lead}} < 1$ – NO assumption on ρ . The $\rho \leq 1$ screening claim is only used to note the further $\times 3$ margin, not for the closure.
- (β) “ $R_{\text{lead}} \leq 0.650$ is the operating-intensity bound; the mid-window (larger A) ratio is larger.” VALID-with-scope: the PHYSICAL competitor sits at the operating endpoint ($A_{\text{op}} \sim 0.013$, small); Math428 separately verified positivity across A for the enumerated readings. The class-wide statement is made at the operating intensity, where $R_{\text{lead}} \leq 0.650$, which is the relevant regime for the Reading-H selection.
- (γ) “Does this close H-diag / flip B1?” DISMISSED: no. It closes the SECOND-CUMULANT off-diagonal stability class-wide; the third cumulant (SC-SCOPE) and the competitor-class definition remain. B1/B2 stay T6 on {H-LAYER}.

Quantitative sanity check (mandatory): *criterion* – band sign = $\text{sign}(1 - \rho R_{\text{lead}})$, verified at products $0.5/1.0/1.5$; *class bound* – $R_{\text{lead}} \leq 23.2 \cdot 14 \cdot 2 \times 10^{-3} = 0.650 < 1$ ($\times 1.54$); *screening* – $\rho = 1/(1+1.03) = 0.493 \leq 1$, product 0.32 ; *consistency* – Math428 enumerated band $+6.7 \times 10^{-4} > 0$ at $\rho = 1$.

7. Result footer

Result ID: B2-PROPA-HLAYER / class-wide second-cumulant off-diagonal stability (rho*R_lead<1)
Precise statement: The off-diagonal Bogoliubov band Delta F_est = Delta F_diag -

$\rho \cdot (1/4) \sum W^2 B$ is positive iff $\rho \cdot R_{\text{lead}} < 1$, $R_{\text{lead}} = [(1/4) \sum W^2 B] / \Delta F_{\text{diag}}$. At $\rho=1$ (no resummation credit) the criterion is $R_{\text{lead}} < 1$; T-018 gives $R_{\text{lead}}(Q) = \text{const} \cdot (1 + K_{\text{floor}}(Q)) \cdot I \leq 23.2 \cdot (1+13) \cdot 2e-3 = 0.650 < 1$ for every admissible competitor ($K_{\text{floor}} \leq T' \leq 13$ pin). Hence $\rho \cdot R_{\text{lead}} \leq 0.650 < 1$ class-wide (margin $\times 1.54$), extending Math428's five enumerated readings (band min $+6.7e-4$ at $\rho=1$) to ALL admissible crystallographic patterns. Screening $\rho=1/(1+g) \leq 1$ ($g=1.03 \geq 0$) improves it (product 0.32).

Scope: Second-cumulant off-diagonal stability; operating endpoint $I=2e-3$; anchor $\mu^2=0.005$. Third cumulant = SC-SCOPE (separate); competitor class = operator item.

Dependencies: hdiag-offdiag-gershgorin-threshold v1.1 (T-018, R_{lead} class bound); offdiag-exchange-scalar-identification (T-019, residual reframing); Math428 (band criterion, ρ , enumerated bands); res-endpoint-admissible-pin ($T' \leq 13$).

Evidence grade: ANALYTIC (band criterion + class bound) + EXECUTED (res5_020_classwide_secondecumulant_stability.py 5/5) + INHERITED (Math428 enumerated bands).

Reproduction command: python codes/vacuum/res5_020_classwide_secondecumulant_stability.py

Expected output: $\rho \cdot R_{\text{lead}}^{\text{class}} (\rho=1) = 0.650 < 1$ (margin $\times 1.54$); screened 0.32; 5/5 PASS.

Falsification gate: An admissible competitor with $R_{\text{lead}} \geq 1$ at the operating intensity (would require $K_{\text{floor}} > T' = 13$, refuted by the pin); or a Math428 band < 0 at $\rho=1$ (refuted: $+6.7e-4$); or anti-screening $\rho > 1$ (would require $g < 0$, an unstable parent).

Tier before / after: No tier flip. B2 T6, B1 T6 on {H-LAYER}. Second-cumulant off-diagonal stability extended from 5 enumerated readings to the whole admissible class.

No-overclaim statement: Closes only the SECOND-CUMULANT off-diagonal stability class-wide. Does NOT close H-diag and does NOT flip B1/B2. The third cumulant (SC-SCOPE, thin) and the competitor-class definition remain. Anchored ($u_{\text{eff}} > 0$, $g \geq 0$ = ROBUSTNESS-MU2).

Next required action: The remaining H-diag residual is the beyond-second-cumulant third cumulant = SC-SCOPE (thin, lifted) extended class-wide, plus the operator competitor-class definition. With this note the A-dependent off-diagonal bubble is class-wide stable.